

Ch. 11, Appendix 1, The Gassmann Equation

The Gassmann equation

Gassmann showed that the compressional velocity, V_p , of the rock could be defined as follows:

$$V_p^2 = \frac{K_s}{\rho} \left(\beta\gamma + \frac{(1-\beta)^2}{B + (1-\beta)} \right)$$

where K = compressibility modulus

ρ = density

$$\gamma = \frac{3(1-\sigma_m)}{1+\sigma_m}$$

σ = Poisson's ratio

$$B = \phi \left(\frac{k_s}{k_f} - 1 \right)$$

ϕ = porosity

$$\beta = \frac{k_m}{k_s}$$

with subscripts s = solid grain material

m = empty matrix (grains + porosity)

f = fluid

Calculation of Matrix Bulk Modulus

Many of the parameters in the Gassmann equation can be defined from logs or estimated from known correlations (see below) but the matrix bulk modulus, K_m , is dependent on the individual rock and must be derived from a known formation prior to fluid replacement. Rearranging the Gassmann equation, K_m can be calculated from the following equation:

$$\beta = \frac{K_m}{K_s} = 1 - A \pm \sqrt{(A+B)^2 - \frac{B^2}{1-\gamma}}$$

$$\text{where } A = \frac{\frac{\rho V_p^2}{K_s} + \gamma(B-1)}{2(1-\gamma)} \quad \text{Note: } V_p \text{ in cm/s !}$$

The value of the bulk modulus derived from this step should be compared with the other moduli to check the consistency of the input data. In all cases the following should be true:

$$K_s > K > K_m > K_f$$

where K is the compressibility of the complete fluid filled rock.

Parameters

The parameters required to evaluate the Gassmann equation are as follows:

- V_p Compressional velocity read from the sonic log
- φ Porosity is usually calculated from the density log, ρ_{log}, using

$$\phi = \frac{\rho_{ma} - \rho_{log}}{\rho_{ma} - \rho_{fl}}$$

using standard values for, ρ_{ma}, matrix density (e.g. 2.65 g/cc in sands, 2.85 g/c in dolomite, 2.71 g/cc in limestone). The fluid density, ρ_{fl}, is defined as follows:

$$\rho_{fl} = S_w \rho_w + (1 - S_w) \rho_{hc}$$

where S_w is the water saturation and ρ_w is the brine density which is dependent on salinity and can be evaluated from chart books or wireline pressure gradients. The hydrocarbon density, ρ_{hc}, can also be evaluated from gradients or from PVT information

- σ_m Poisson's ratio for the matrix is best calculated from properly QC'd shear logs where these have been recorded

$$\sigma = \frac{V_s^2 - \frac{1}{2} V_p^2}{V_s^2 - V_p^2}$$

but in the absence of shear logs, data can be estimated for sandstones from one of the following correlations:

$$\begin{aligned} \sigma_m &= 0.267 - 0.65\phi && \text{(consolidated sands)} \\ \sigma_m &= 0.214 && \text{(unconsolidated sands)} \end{aligned}$$

In the absence of other data a value of 0.25 is normally taken, but in any case 0 < σ_m < 0.5.

- K_s The grain bulk moduli are typically taken from standard references as follows:

Sandstone 38 GPa
Limestone 82 GPa
Dolomite 67 GPa

- K_f The fluid bulk modulus depends on the fluid type. A typical value for water/brine is 3.1 GPa but this is strongly dependent on salinity, temperature and pressure and values should be derived from available references for all fluid moduli, including oil and gas (see References, especially *Batzle & Wang*). The PE application PVTSYM is also useful in predicting changes with temperature and pressure.

Note that the Gassmann equation is theoretically only valid for single phase fluids and that mixed fluids are assumed to behave as homogenous fluids of the same density. Wood's formula is used to calculate the resultant bulk modulus e.g for gas:

$$\frac{1}{K_f} = \frac{S_w}{K_w} + \frac{1 - S_w}{K_g}$$

Information on fluid compressibility may be also be available from PVT analysis ($K=1/C$)

Fluid Substitution

Once K_m has been defined, the assumption is made that changing the fluid content has no effect on K_m , K_s , σ_m or ϕ , and the original Gassmann equation is then used with revised values of K_f and ρ to calculate the new compressional velocity when the fluid content is varied.

Gassmann Fluid Replacement Step by Step

- 1) Determine values for the bulk moduli of the solid grain material, K_s , and the fluid, K_f , as indicated above.
- 2) Determine the porosity, ϕ , from the density log using the appropriate matrix density, ρ_{ma} , and fluid density, ρ_{fl} , calculated from the known fluid make-up.
- 3) Evaluate the Gassmann parameters B and γ .
- 4) Evaluate A and β . Note that the selected root for β must be between 0 and 1 in order to ensure that $K_m < K_s$.
- 5) Calculate a revised ρ_{fl} for the new fluid content and a new bulk modulus, K_f .
- 6) Calculate the new value of log density and B and substitute into the Gassmann equation assuming that none of β , σ_m or ϕ are dependent on fluid properties.

Units

In order for the Gassmann equation to be used properly the units used must be consistent. In manual calculation, problems are best avoided if the following units are employed:

Bulk modulus	dynes/cm ²
Density	g/cc
Velocity	cm/s
Porosity	fract

For convenience the following conversions can be used:

ft/s	ρ	m/s	$\times 0.3048$
$\times 10^{10}$ dynes/cm ²	ρ	GPa	$\times 1.0$
g/cc	ρ	kg/m ³	$\times 1000$

Gassmann's Assumptions

In deriving his original equations, Gassmann made a number of assumptions which should be borne in mind when carrying out fluid replacement:

- 1) All pores are connected and the fluid in the pore space is non-viscous i.e. the fluid is in hydrostatic equilibrium at all times.
- 2) Hooke's law holds for all the components in the system (grain, matrix, fluid, fluid-filled rock):

$$\frac{\Delta V}{V} = -\frac{\Delta P}{K}$$

- 3) The rock is homogeneous and isotropic on a macroscopic scale.
- 4) The solid grain material is homogeneous and isotropic.
- 5) The shear modulus, μ , is independent of fluid content i.e. $\mu = \mu_m$

It is also worth noting that fluid substitution is only valid into 100% lithologies. Accounting for mixed lithologies such as shaly sands would require a more detailed approach.

Graphically the effect of gas and oil on V_p and ρ is shown in Figure A1.

Related Equations

The theoretical work behind the Gassmann equation utilise a number of other relations some of which can be useful in determining input parameters for different applications.

Compressional velocity $V_p = \sqrt{\frac{M}{\rho}} = \sqrt{\frac{K + \frac{4}{3}\mu}{\rho}}$

$$M = K + \frac{4}{3}\mu$$

Shear velocity $V_s = \sqrt{\frac{\mu}{\rho}}$

Poisson's ratio $\sigma = \frac{3K - M}{3K + M} = \frac{V_s^2 - \frac{1}{2}V_p^2}{V_s^2 - V_p^2}$

Compressibilities $K_s = \frac{1 + \sigma_s}{3(1 - \sigma_s)} \rho_s V_{p,s}^2$

$$K_f = \rho_f V_{p,f}^2$$

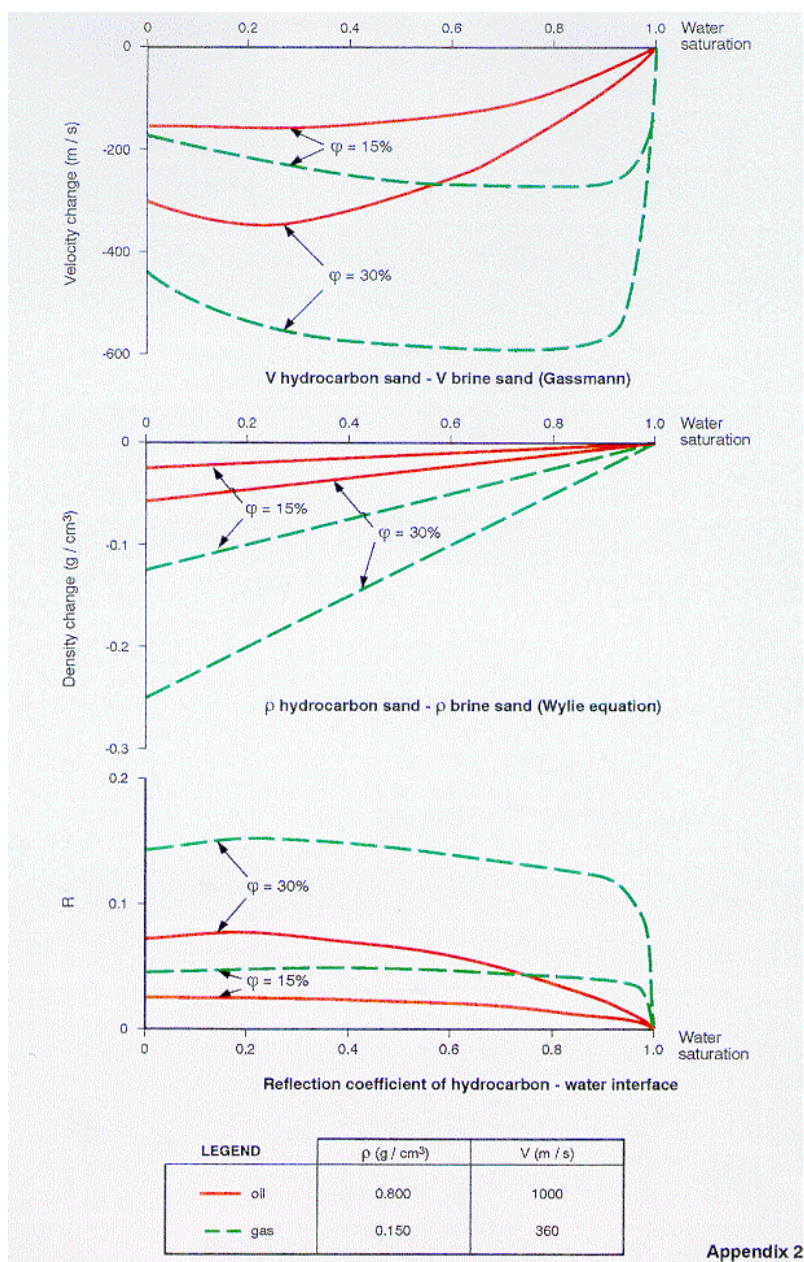


Figure A1 - The Effect of Porefill on V_p and ρ